



3.3 Differentiating Inverse Functions

Definitions of Inverse Functions

Two functions are inverse functions when

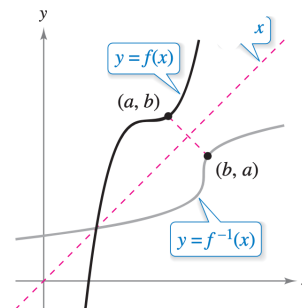
$$(f \circ f^{-1})(x) = \underline{\hspace{2cm}}$$

AND

$$(f^{-1} \circ f)(x) = \underline{\hspace{2cm}}$$

Inverse Properties

- The graphs of f and f^{-1} are reflections about the line $y = x$.
- The domain of f is the range of f^{-1} .
- The range of f is the domain of f^{-1} .
- A function has an inverse function if and only if it is one-to-one, that is its monotonically increasing or decreasing.



Ex 1: Find the indicated value.

a) $f(x) = x^{5/3}$, $f^{-1}(243) =$

b) $f(x) = 2x + 7$, $f^{-1}(14) =$

Discovering Relationships

Ex 2: Open the [GeoGebra applet](#). Move slider a and observe the values of f' and g' . What do you notice about the slopes $f'(a)$ and $g'(b)$?



The Derivative of an Inverse Function

Let f be a function that is differentiable on an interval I and has an inverse function f^{-1} . If f is differentiable at on an interval containing $x = c$ and $f'(c) \neq 0$, then f^{-1} is differentiable at $x = f^{-1}(c)$. Moreover,

$$(f^{-1})'(x) = \underline{\hspace{2cm}}$$

In words $(f^{-1})'(x)$ is pronounced the derivative of the inverse function of f at x .



On the AP exam, inverse functions are often not expressed using inverse notation. For example, let f be a function with inverse function g then $g'(x) = \underline{\hspace{2cm}}$.



Ex 3: Find the indicated value.

a) $f(x) = x^{5/3}$, $(f^{-1})'(243) =$

b) $f(x) = 2x + 7$, $(f^{-1})'(14) =$

Ex 4: If $f(x) = x^5 + 2x^3 + x - 1$ and $(f \circ g)(x) = (g \circ f)(x) = x$, find $g'(3)$.

Ex 5: If $f(x) = x^3 + x$ and $h(x)$ is the inverse of $f(x)$, then $h'(2)$ is

- A) $\frac{1}{13}$
- B) $\frac{1}{4}$
- C) 1
- D) 4
- E) 13

SCORING STATS!

Guess the percentage of students
who answered correctly:
____%

Ex 6: Let f be a differentiable function such that $f(3) = 15$, $f(6) = 3$, $f'(3) = -8$ and $f'(6) = -2$. The function g is differentiable and $g(x) = f^{-1}(x)$ for all x . What is the value of $g'(3)$?

- A) $-\frac{1}{2}$
- B) $-\frac{1}{8}$
- C) $\frac{1}{6}$
- D) $\frac{1}{3}$
- E) The value of $g'(3)$ cannot be determined from the given information.

SCORING STATS!

Guess the percentage of students
who answered correctly:
____%



Ex 7: 2007 AB3 – part (d)

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

SCORING STATS!

Guess the mean: _____/9

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x . The function h is given by $h(x) = f(g(x)) - 6$.

- (d) If g^{-1} is the inverse function of g , write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.

Ex 8: Spicy 🌶️🌶️🌶️

Suppose g is the inverse function of a differentiable function f and $G(x) = \frac{1}{g(x)}$. If $f(3) = 7$

and $f'(3) = \frac{1}{9}$, find $G'(7)$.

- (A) -5 (B) 4 (C) 6 (D) -1 (E) -4

**This problem was authored by Calculus Maximus*